

The Electoral Costs of Within-Bloc Competition: A Nationwide Canvassing Experiment

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Abstract

In proportional systems with electoral thresholds, competition within an ideological bloc can mechanically reduce its representation by splitting support across party lists. We examine whether within-bloc competition also changes voter behavior through a nationwide randomized canvassing experiment with the Social Democratic Party in Croatia’s 2024 parliamentary election. The party’s coalition talks with the Greens failed, and the parties ran separate campaigns. Conditional on the Greens campaigning in a precinct, we estimate the effect of randomly assigned Social Democratic canvassing. The random assignment varies one component of within-bloc competition for the same voters. To determine where the Greens campaigned, we geolocate more than 400 campaign photographs posted on social media, matching visible features in each image to Google Street View. In these areas, Social Democratic canvassing increases the incumbent Christian Democrats’ vote share by 0.6–1.0 percentage points and reduces turnout by a similar amount. The results identify an additional cost of within-bloc competition through voter demobilization.

1 Introduction

Coalitions have been a central feature of proportional representation (PR) systems throughout the history of developed democracies. Over recent decades, however, the parties that dominated 20th-century European politics—particularly mainstream center-left and center-right parties—have experienced a steady decline, coinciding with the rise of new challenger parties across the ideological spectrum (De Vries and Hobolt, 2020). The increasingly fragmented political environment heightens the importance of coalitions in government formation and their role in presenting voters with clear and viable policy alternatives (Golder Nadenichek, 2006).

Recent years have seen several high-profile attempts to form pre-electoral coalitions, particularly aimed at countering the rise of right-wing populist and far-right parties. In Poland, the Civic coalition successfully unseated the right-wing populist Party PiS in the 2023 parliamentary election.¹ In contrast, in Hungary the big-tent coalition United for Hungary failed to defeat Viktor Orban’s Fidesz party in the 2022 parliamentary election.² In the Netherlands, the Labour Party (PvdA) and the Greens (GroenLinks) have formed a lasting alliance, competing with a joint list in the 2023 parliamentary election, where Geert Wilders’ far-right Pvd party won the largest number of votes (van Holsteyn and Irwin, 2025). The two parties are likely to fully merge into one.³

The emergence of “niche” parties that may be unwilling or unable to effectively moderate or compromise on their policy positions (Adams et al., 2006) often complicates coalition-building in systems with relatively high effective thresholds for representation.

Evidently, the coalitions that have been formed have had mixed success in terms winning elections, but we do not know how parties would have performed in the absence of the pre-electoral coalitions (and vice-versa if no pre-electoral coalition was formed). It is not clear how voters respond to pre-electoral coalitions, whether coalitions are strategically beneficial for the parties that enter them (in the short or long term), and what consequences that has for legislative representation in PR systems.

Pre-electoral coalitions can affect electoral outcomes through multiple channels. Running independently allows parties to maintain programmatic distinctiveness or avoid compromising on their platform, while coalition-building can offer mechanical advantages under many electoral systems (Golder Nadenichek, 2006) and may also signal competence to voters, especially consequential in a pre-electoral campaign context. Importantly, prior research indicates that campaign activity, including relatively simple signals such as candidates campaigning may shape voter beliefs about candidate valence (Barton, Castillo and Petrie, 2014). Which of these dynamics – (i) voters’ preferences for purity in their chosen party’s platform or (ii) the signal that the parties can reach an agreement, are capable of governing together and are more likely to win because of electoral system disproportionality – prevails remains an open empirical question. Importantly, these two countervailing dynamics exist regardless of whether a coalition is formed, but they point in opposite directions depending on the coalition decision. If a coalition is formed, voters who prefer ideological purity may be discouraged from supporting the coalition, while voters who view coalition-building as a signal of competence—and recognize that larger, unified electoral blocs are more likely to succeed

¹<https://researchbriefings.files.parliament.uk/documents/CBP-9951/CBP-9951.pdf>

²<https://researchbriefings.files.parliament.uk/documents/CBP-9519/CBP-9519.pdf>

³<https://nos.nl/artikel/2559385-fusie-groenlinks-en-pvda-komt-in-stroomversnelling-besluit-met-jaar-vervroegd>

due to mechanical advantages built into the electoral system—may be more likely to support the coalition. Conversely, if a coalition is **not** formed, voters valuing programmatic purity retain the ability to choose the individual party more closely aligned with their ideological preferences, but those who would have been mobilized by a credible signal of competence and increased electoral viability under the electoral system may be less likely to turn out or lend their support. We examine the second case and contribute new causal evidence from a real-world field experiment that captures how voters respond to a failed coalition agreement between two ideologically aligned parties. While it is possible that the total effect (including mobilizing and demobilizing dynamics) is not symmetrical between the coalition and no-coalition case, we believe it is important to document that a demobilizing effect exists also if there is no coalition.

We study this question in the context of the Croatian 2024 parliamentary elections. In Croatia, the PR electoral system, combined with the D’Hondt apportionment method, produces high thresholds for legislative representation and therefore strong incentives to form coalitions. Ahead of the election, the Social Democrats (SDP) and the Greens engaged in extensive negotiations over forming a joint list, but ultimately failed to reach an agreement and the green party Mozemo decided to compete on their own.⁴

We use the data from a nation-wide canvassing experiment conducted by the Social Democratic Party (the second largest national party both before and after the election) in Croatia ahead of the 2024 parliamentary election. The experiment was suggested and designed by us as researchers and implemented by the party. The treatment, administered by party members and volunteers, included a brief message encouraging voters to vote for the Social Democratic Party of Croatia at the election and a leaflet highlighting the party’s policies and candidates.

The experiment involved 1716 precincts which covered 1.4 million people, around 40% of Croatia’s voting population. We use administrative data at the granular precinct level, where one precinct on average covers around 800 people. Precincts were randomly assigned to either treatment or control, with 80% in the former group and 20% in the latter. In addition to examining the Social Democrats’ randomized campaign activities, we obtain data on the (non-randomized) canvassing locations of the Greens using Open Source Intelligence (OSINT) geolocation methods. This allows us to estimate the effect of the Social Democrats trying to win over the same voters as the Greens, despite previously considering creating a joint list which would have appealed to voters of both parties. Specifically, we estimate the effect of the Social Democrats’ randomized campaign, conditional on the Greens campaigning in that area – meaning that we restrict our sample to the voting booths ("precincts") where the

⁴<https://vijesti.hrt.hr/hrvatska/mozemo-sdp-10863643>

Greens conducted their campaign. This identifies the treatment effect among voters exposed to the Greens but does not inform how voters in areas the Greens did not target—likely less receptive to the progressive party—would respond to the treatment.

The randomized canvassing experiment likely informs some voters that the coalition agreement was not reached – Stevenson and Vavreck (2000), Arceneaux (2005) and Kendall et al. (2015) demonstrate that voters learn from campaigns, which can extend to learning about the characteristics of the *competitor*, not just updating beliefs about the campaign that delivered the message. The campaign’s role — including canvassing — in delivering the message that the two parties were competing separately may have been especially influential given that the coalition negotiations collapsed less than four weeks before election day⁵. An alternative and weaker assumption is that canvassing by two separate parties simply raises the salience of the two parties competing against each other instead of forming a joint list.

We find that when the two prospective coalition partners compete for the same voters through the use of canvassing, the main rival – the incumbent Christian Democratic party (HDZ)– benefits instead, as their vote share increases by 0.6 - 1 percentage points. Turnout effects are less precisely estimated but are directionally consistent with a demobilization mechanism: in our main specifications, turnout declines by 0.6 to 0.8 percentage points, closely mirroring the size of HDZ’s vote share gains. This pattern suggests that the competition between coalition partners primarily depressed participation among left-leaning voters rather than shifting partisan preferences toward HDZ. The consistently negative, though sometimes imprecisely estimated across different specifications, turnout effects are also inconsistent with the interpretation that exposure to two left-wing campaigns galvanized right-leaning voters.

The finding has implications for strategic behavior of coalition partners in multi-party, fragmented electoral systems and for the legislative representation of voters whose ideological camp is divided across multiple parties. The experiment suggests that some voters are demobilized by the failure of ideologically aligned parties to cooperate—perhaps because they perceive the fragmented political block as unlikely to succeed. While there may also exist expressive voters who reject compromise on their preferred party’s platform, our findings indicate there exists another group of voters who are demobilized by the *lack* of compromise and diminished prospects for winning more seats. These voters may be likely less tightly affiliated with individual parties and may not be captured in the typical instruments used for strategic decision-making, such as polls of party members⁶.

⁵<https://www.jutarnji.hr/vijesti/hrvatska/propali-pregovori-izmedu-sdp-a-i-mozemo-imamo-prve-detanje-o-raspadu-tockaste-koalicije-15442091>

⁶<https://www.nrc.nl/nieuws/2023/07/18/linkse-samenwerking-zet-historische-stap-maar-is-nog-niet-voltooid-a4170027>

2 Literature Review

This paper contributes to three main strands of research. It engages with the literature on electoral campaign interventions by focusing on the electoral effects of canvassing in a multi-party context and the campaign *actions* of more than one party. It also expands on existing studies of voter responses to pre-electoral coalitions, providing empirical evidence from a large-scale experiment conducted by a real-life political party. Finally, to our knowledge, this is the first social science paper to employ open-source investigative methods (OSINT).

A large body of research studies how campaigns mobilize and persuade voters. Gerber and Green (2000, 2001) show door-to-door canvassing raises turnout, while phone and mail have limited impact. Bergan et al. (2005), Nickerson (2008), Nickerson, Friedrichs, and King (2006), and Enos, Fowler, and Vavreck (2014) explore grassroots mobilization, household spillovers, habit formation, outreach methods, and turnout disparities. Other studies focus on persuasion. Arceneaux and Nickerson (2009), Dewan, Humphreys, and Rubenson (2014), Barton, Castillo, and Petrie (2014), and Cantoni and Pons (2020) examine message tone, endorsements, personal contact, and candidate visits. Cruz et al. (2024) and Broockman and Kalla (2023) show that voters respond to substantive campaign content—whether through programmatic policy promises in clientelist settings or structured conversations that reduce exclusionary attitudes.

Kendall, Nannicini, and Trebbi (2015) show that messages emphasizing valence (e.g., competence) rather than ideology significantly shift voting behavior and shape beliefs about both the candidate and their opponents. This empirical insight aligns with formal theoretical accounts that emphasize the electoral importance of perceived competence (e.g., Bernhard et al., 2010; Galasso and Nannicini, 2011; Groseclose, 2001). The issue of competence is particularly salient in this project’s context, as challenger parties—especially Green parties—often struggle to be seen as credible governing alternatives (Abou-Chadi and Kamphorst, 2023; Zur, 2019). Pons (2018) contributes nationwide field experiment on partisan canvassing in France, demonstrating that direct contact affects vote shares due to persuasion in a high-stakes, multiparty presidential election.

The closely related literature on campaign advertising finds modest but consistent persuasive effects. Ansolabehere and Iyengar (1995) link negative ads to demobilization. Huber and Arceneaux (2007) use variation in TV ad exposure to document persuasive effects. Large-scale work by Coppock, Hill, and Vavreck (2020) and Hewitt et al. (2024) confirms small but real average effects, often maximized by the wealthier campaigns able to test and scale content. Spenkuch and Toniatti (2018) also exploit discontinuities in media market boundaries to show that Democratic and Republican TV ads largely offset each other. It is a rare study

that analyzes the interactive effects of multiple (advertising) campaigns, albeit in a two-party setting.

Finally, a few studies examine campaign dynamics in multiparty settings. Galasso, Nannicini, and Nunnari (2021) show that negative advertising in a three-candidate race produces backlash and spillovers to uninvolved competitors. However, they do not examine the actions of more than one campaign. Esteban Casanellas (2024) offers the only randomized field study examining competing party actions, showing that overlapping street-level *posters* can measurably shift support. Much like Galasso, Nannicini, and Nunnari (2021), we find that in multiparty settings, competing campaign activity—even when the campaign tone is neutral—can benefit third parties. However, our study is the first to analyze electoral effects of multi-campaign canvassing, capturing competition between parties and the role of potential coalitions in shaping vote choice.

While there is a substantial literature on coalition formation and their behavior in the post-electoral context (see Laver 1998 for a review), pre-electoral coalitions (PECs) operate under different incentives and may be perceived quite differently by voters. Golder Nadenichek (2006) highlights the prominence of PECs in the postwar period—often formed strategically, and when absent, still a product of deliberate calculation. Ibenskas (2016) provide additional descriptive evidence on pre-electoral coalition formation and frame it as a rational response to electoral incentives—more common under disproportional or fragmented systems, and often driven by ideological fit and expected gains.

Yet despite their importance, PECs remain comparatively understudied through high-quality empirical approaches. A collection of studies examines how coalition signals shape voter perceptions, though much of this evidence is drawn from surveys or laboratory experiments, some with student samples. Duch et al. (2010) show voters factor expected coalitions into vote choice. Gschwend, Meffert, and Stoetzer (2017) and Bahnsen et al. (2021) find coalition cues affect beliefs about governing outcomes. Falco-Gimeno and Muñoz (2017) show such cues shift perceived party ideology. Gschwend and Hooghe (2008) and Meffert and Gschwend (2010) find these effects are moderated by ideological fit, partner appeal, and party size.

Recently, stronger evidence has emerged from structural models and quasi-experimental approaches. Montero (2024), using a structural model, finds that pre-electoral coalitions in Mexico lower campaign costs and benefit participating parties electorally. Frey et al. (2023) show that ideologically distant parties sometimes form PECs to dislodge entrenched incumbents, again using structural methods applied to Mexican data. Hortala-Vallve et al. (2023) study *local* elections in Finland and provide descriptive and evidence that PECs are more likely between similarly sized, ideologically proximate parties, particularly in disproportional systems. Using difference-in-difference methods, they show that voters punish

coalescing parties and strategically reallocate personal votes within coalitions.

Existing evidence points in different directions, with limited consensus across a small but growing body of research; we contribute new evidence to this literature through a nation-wide experiment in a high-stakes election where a pre-electoral coalition could have been politically very consequential. We find that the absence of coordination among ideologically aligned parties can reduce turnout and advantage parties outside the potential coalition bloc.

Finally, we are the first, to our knowledge, to apply OSINT methods in the social sciences. We construct a dataset of competitors’ campaign activities by geolocating social media images through a rigorous investigative process matching photos to exact coordinates. Unlike prior work relying on user-provided geo-tagged metadata (e.g., Barrie et al. 2023; Fujiwara et al. 2024), our approach adapts techniques employed by investigative journalists. While our manual process achieved a 95% geolocation success rate, advances in large language models could automate key steps and scale this method. The vast volume of publicly available imagery offers a promising and largely untapped resource for research in political economy.

3 Pre-electoral Coalitions: the Theory and how canvassing experiments can help them elucidate the dynamics at play

Pre-electoral campaigns are expected to (i) inform voters about party platforms and (ii) influence voters’ evaluations of party valence.

In certain electoral systems, such as PR with the D’Hondt apportionment method (a common setting in European democracies), forming a pre-electoral coalition carries benefits. This is because D’Hondt favours larger parties, but also because it reduces the likelihood of a party staying below the threshold for representation in a particular district.

In a setting where two parties can form a pre-electoral coalition, the pre-electoral campaign carries an additional signal—it communicates that either two parties now have a joint platform (affecting channel (i)) and they are able to reach an agreement and govern together, should they win (affecting channel (ii)), or that, conversely, their platforms differ (i) and they are not able to reach an agreement and govern jointly (ii). Therefore, in a case where pre-electoral coalitions are possible, the basic decisions voters must make still concern congruence with their policy preferences and the ability of a given party to implement them, but the ability to reach a coalition deal is a new crucial dimension of the voters’ evaluation of party valence.

We study a setting where two ideologically similar parties attempted, but failed, to reach a pre-electoral coalition agreement. In order to identify voter responses to the two parties campaigning separately, we utilize a campaign experiment conducted by the Social Democrats, who randomized the locations where they canvassed. We also construct a dataset

of the campaign locations of their main left-wing competitor, the Green Party (Mozemo). This allows us to estimate the effect of the Social Democrats' campaign, conditional on Mozemo's decision to campaign in a given area. In treated units, both parties campaign; in control units, only Mozemo does. We expect that voters who are exposed to both the Greens' and the Social Democrats' campaign will learn that the platforms are very similar and that the two parties were not able to reach an agreement.

While the treatment does not randomize the actual existence of a pre-electoral campaign, given the media attention regarding coalition negotiations preceding the campaign, the fact that the two parties are *not* campaigning jointly – a signal of the parties' competence – may provide the most consequential new information for voters, who may be familiar with the parties' general positions. The effects of canvassing are generally small, but the treatment does still either carry new information about the parties for some voters, or at least, it raises the salience of the fact that there is no coalition.

While there may exist cases where the platforms have significant differences (as in the case of Frey et al. (2023), a standard result in the literature is that pre-electoral coalitions are much more likely between ideologically similar parties (Golder Nadenichek, 2006)). In this case, local academic observers⁷ have classified the two parties as quite similar on both the social and (even more so) the economic dimension of their platforms. Both parties' also emphasized similar policies during the campaign, such as the greater direct government provision of affordable housing^{8,9}. This means that, in this case, the voters may learn that there would have been no major platform tradeoffs from forming a coalition, and that the inability to form one makes it less likely the parties are able to win (that is, gain the largest number of seats) and govern effectively. We expect that the fact that election is less close in the absence of a coalition will induce some voters to abstain from voting, consistent with the canonical calculus of voting models (Downs, 1957; Tullock, 1968) or related formulations of the pivotal voter model (Ledyard, 1984; Palfrey and Rosenthal, 1985), as well as the calculus voting model (Riker and Ordeshook (1968)). It is also consistent with more recent modifications incorporating social as well as selfish preferences such as Edlin et al. (2007) or alternatives such as Feddersen and Sandroni (2006), which both predict higher turnout in close elections – itself subject to a large literature, including by Burzstyn et al. (2024) – and can explain empirically observed turnout levels in large elections.

⁷<https://www.index.hr/vijesti/clanak/provjerite-jeste-li-desno-ili-lijevo-i-koja-stranka-vam-je-najbliza/2553905.aspx>

⁸<https://mozemo.hr/predstavili-smo-stambenu-politiku-mozemo>

⁹<https://www.facebook.com/sdphrvatske/videos/364519383220007>

4 Setting: Croatia's 2024 Parliamentary Election

Croatia employs a party-list proportional representation system with D'Hondt apportionment. The parliament comprises 151 members: 140 elected from ten multi-member constituencies (14 seats each), plus 3 seats for the diaspora and 8 seats for ethnic minorities. Approximately 3.5 million citizens are eligible to vote. The 2024 parliamentary election was called on March 15 and held on April 17, earlier than anticipated. The incumbent party, the Christian-democratic HDZ, has been in power since 2015.

Before the election, the two main left-wing parties, SDP and Mozemo, tried to form a coalition. Despite the aforementioned programmatic similarities, negotiations failed. The parties also tried to negotiate a partial coalition – i.e. create a joint list in only some of the districts, but not the entire country. The negotiations focused on four districts (out of 10 in total) where Mozemo was in peril of not exceeding the threshold for parliamentary representation, according to polls. A significant point of contention was the ranking of each party's candidates on potential shared (coalition) lists in those four districts, determining which individuals would be likely to secure parliamentary seats.¹⁰ Mozemo was offered two certain positions on the joint lists and, as it believed it would win four individually, it decided not to form the coalition with SDP.¹¹

HDZ secured 34.42% of the vote, obtaining 61 seats. They formed a post-election coalition with the far-right Homeland Movement (DP), which received 9.56% of the vote and 14 seats. This alliance enabled HDZ to maintain a governing majority after being backed by the 8 MPs representing ethnic minorities, who normally side with the biggest party in the election.

SDP won 25.4% of the national vote (42 seats), while Mozemo garnered 9.10% , translating to 10 seats. Mozemo failed to surpass the electoral threshold in four out of ten districts (3 of which were subject to "partial" coalition negotiations), resulting in no representation from those areas and "wasted" votes for those voters. A coalition between SDP and Mozemo could have mechanically yielded six additional seats for the bloc. This relies on keeping all vote shares constant and simply adding up the two parties' vote share – the full calculation is in the Appendix. The 6 additional seats the left-wing coalition would have won (keeping voter behavior constant) would have cost HDZ three seats and the far-right another three. This would result in the left-wing coalition winning the largest vote share (34.5%) and the same number of seats as HDZ, placing them in a strong position to lead government formation talks.

¹⁰<https://net.hr/danas/vijesti/propali-pregovori-izmedu-sdp-a-i-mozemo-nema-tockaste-koalicije-b088b3ac-e86a-11ee-9f94-7ecc6c482c09>

¹¹<https://www.telegram.hr/politika-kriminal/u-sdp-u-dosta-furiozno-docekali-odluku-mozemo-ako-je-nekome-matematika-teska-kao-paprika-njihov-problem/>

Following the election, Mozemo leader and Zagreb Mayor Tomislav Tomašević expressed regret over the failure to form even a partial coalition with SDP.¹² Subsequently, in the lead-up to the 2025 local elections, SDP and Mozemo formed a coalition in Zagreb, with Tomašević as their joint mayoral candidate.¹³ The coalition is expected to win convincingly.¹⁴

5 Data

We use precinct-level data on turnout and vote shares, covering 1716 in the experiment and over 6,000 nationally. The list of precincts included in the experiment was generated using data from the 2020 election. Because of relatively minor precinct boundary changes whose precise shape was unclear until late in the campaign, we can observe the outcome for 1666 precincts that we can reliably match between 2020 and 2024. This is a high match rate, as for earlier elections a much lower percentage of precincts can be matched with certainty. The experiment included precincts populated by 1.39 million Croatian voters – 39% of the voting population. To identify the effect of SDP campaigning conditional on Mozemo’s presence, in the main specification, we restrict the analysis to areas where Mozemo canvassed. This allows us to estimate the campaign’s impact on the performance of the two parties, as well as others, in areas Mozemo was active. From conversations with party members, we know that Mozemo positively selects the areas it campaigns in, based on their own past performance and, even more importantly, busy-ness of the location. Our estimate of the effect of SDP’s campaign holds constant the selection Mozemo engages in. However, this reduces external validity, as it limits the scope to locations where Mozemo has chosen to campaign, which may not be representative of all areas in the country. Our data sources include internal records from SDP on canvassing activities and open source intelligence, i.e. social media content. Mozemo places significant emphasis on digital outreach and frequently announces campaign locations on social media ahead of time, in addition to documenting areas they had visited. We collected 206 posts and 438 images across major Mozemo accounts—national, regional, and affiliated canvasser or politician accounts—on Instagram and Facebook, the dominant platforms in Croatia. This process yielded 166 unique campaign locations based on coordinates.

¹²<https://www.tportal.hr/vijesti/clanak/tomasevic-priznao-zao-nam-je-sto-nismo-isli-u-tockastu-koaliciju-sa-sdp-om-20240420>

¹³<https://www.tportal.hr/vijesti/clanak/sdp-i-mozemo-zajedno-na-izborima-u-zagrebu-njihovi-protukandidati-nisu-stedili-rijeci-foto-20250301>

¹⁴<https://vijesti.hrt.hr/hrvatska/hrejting-16-12147337>

6 Geolocating Mozemo’s Campaign Activities

We geolocated over 400 campaign-related images, identifying 166 unique locations. Many images captured the same place from slightly different angles or times, allowing us to triangulate locations through visual overlap and contextual clues. In total, 95.2% of locations were identified with high certainty, and the resulting distribution closely matches anecdotal lists of campaign locations provided by Mozemo canvassers. The process deployed is a form of open-source intelligence (OSINT), commonly used by investigative journalists, including a Pulitzer Prize-winning team at The New York Times. One high-profile case of OSINT application was the identification of Russian opposition leader Alexei Navalny’s would-be assassins, conducted by the organization Bellingcat in collaboration with Navalny¹⁵.



Figure 1: Mozemo’s original Instagram post from Ivanić-Grad

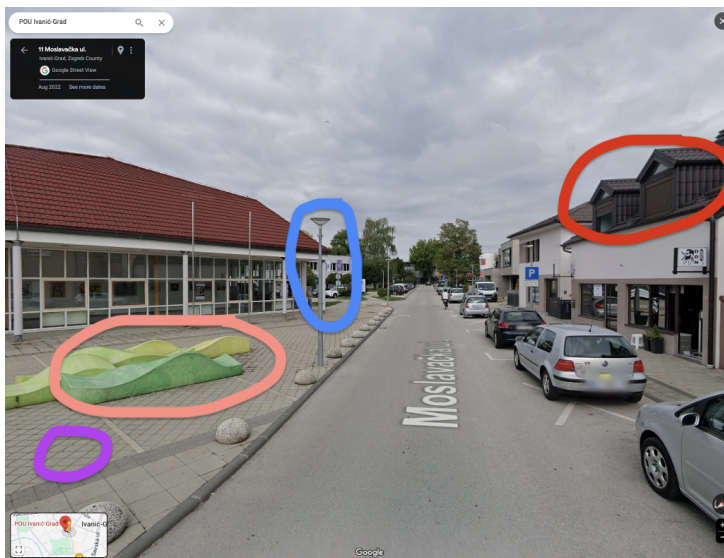


Figure 2: Geolocated site of Mozemo’s post in Ivanić-Grad

Our goal was to match each image to its precise location, ideally replicating the original camera frame. This involved identifying stable visual cues—storefronts, signage, architectural features, and even tree lines. Once we knew the general region or a list of potential towns or

¹⁵<https://www.newyorker.com/news/dispatch/how-bellingcat-unmasked-putins-assassins>

cities (which was often provided in the social media posts), commercial signage often played a key role: supermarket chains or business have recognizable logos, enabling us to search known locations and verify them using Google Street View. We can confirm whether there is a match by aligning facade colors, signage placement, nearby buildings, etc. Figures 1 and 2, as well as 3 and 4, provide examples of geolocated sites of Mozemo’s campaign. The first example, in Figures 1 and 2, was easier to identify, as Mozemo announced on Instagram, within the image, that the campaign was taking place in the town of Ivanić-Grad. The second example (Figures 3 and 4) also included reference to the general area – the Verudela neighborhood of the coastal town of Pula. However, Verudela includes a very long, tree-lined walk along the coastline, which is where the campaign took place. The absence of buildings creates a challenge, but we were able to geolocate the exact site of the campaign, as evidenced by the patterns in the image, including the shape of the tree, the beach and the sidewalk. While we identified 166 campaign locations, the campaign deliberately chose busy areas with



Figure 3: Mozemo’s original Instagram post from Verudela



Figure 4: Geolocated site of Mozemo’s post in Verudela

a lot of foot traffic (often squares, local markets, etc.). This suggests substantial spatial spillovers, with exposure diminishing with distance. Drawing a circle around the location is a reasonable approach to identify overlaps between the two campaigns. This also means that many more *precincts* are covered by Mozemo campaigns as one set of coordinates covers multiple precincts. On the other hand, some locations where Mozemo campaigned were not included in the experimental randomization at all, in which case they are excluded from our dataset.

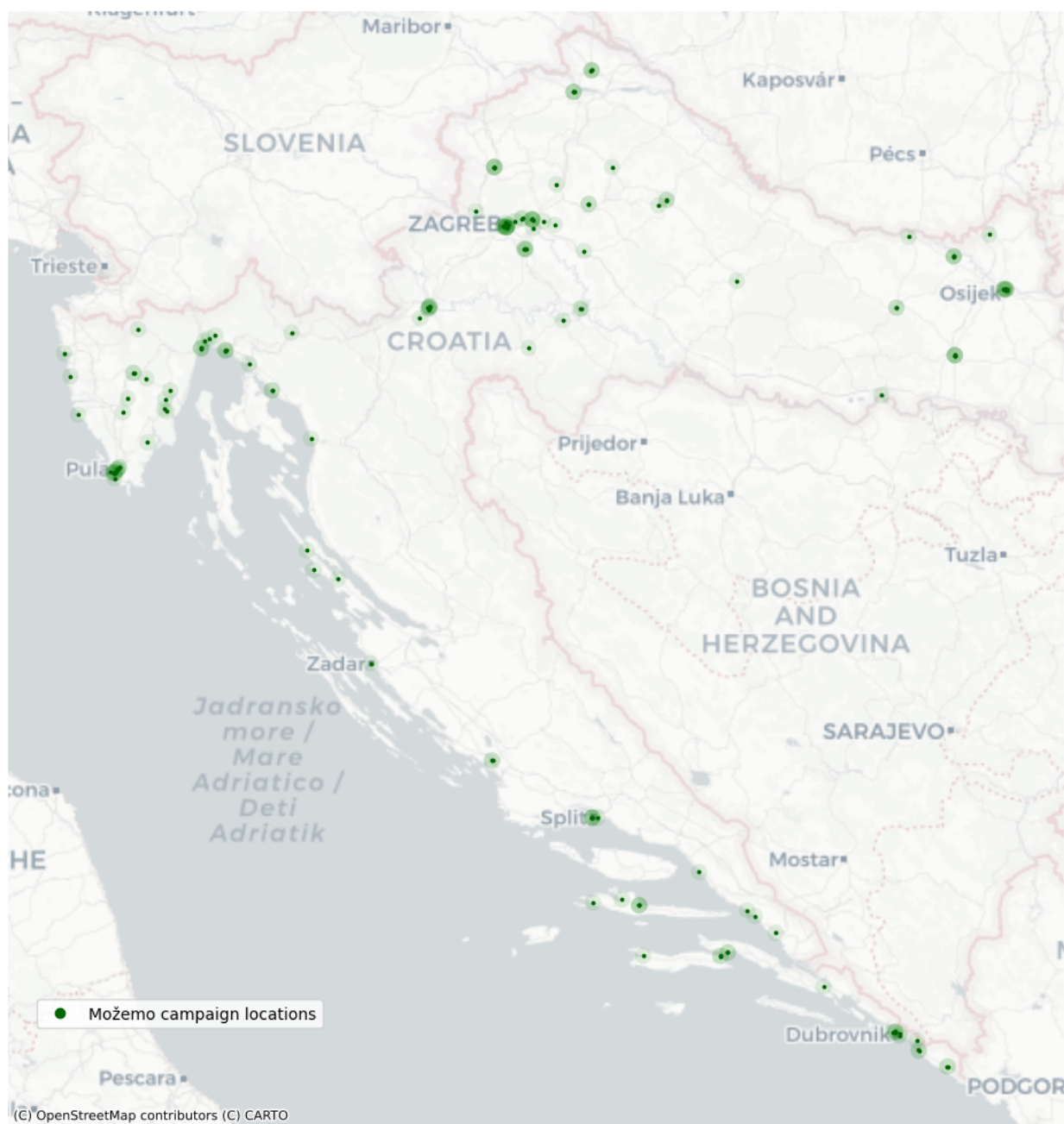


Figure 5: Geolocated Možemo campaign activities with 5 km radii around them for greater visibility

7 Creating Maps of District Boundaries

Precincts in Croatia, identified as “voting booths”, are quite small — each covers 836 people on average within our sample. However, no official shapefiles exist for them. One specific challenge this presents is that a single voting booth address (e.g., a school) might host multiple precincts (e.g., booths 35 and 36), each serving different geographic areas and potentially assigned to different experimental conditions. To construct adequate spatial representations of these precincts, we relied on the Google Maps API, and identified the coordinates of each range of street numbers covered by a single precinct (which often comprises subparts of several streets). This method was necessary to ensure precision in defining the spatial boundaries of treatment and control areas and determining the overlaps with Mozemo’s campaign. Because precincts were defined based on (sometimes changing) sets of addresses and not stable geographic units, matching them across years posed additional challenges. While we were able to match 97% precincts between the 2020 and 2024 elections, the quality of matches deteriorates substantially for earlier elections, with only 57% of 2016 precincts successfully matched. For this reason, earlier results are not used in the analyses below as they significantly reduce and skew the sample.

8 Experimental Setup

The Social Democrats’ campaign was implemented across 1,361 treatment precincts and 355 control precincts in 10 districts, with randomization conducted at the district level. 80% of precincts within each district were allocated to the treatment and 20% were allocated to the control group. Assuming a 3.2 percentage point effect, as in Pons (2018), we may encounter issues with statistical power. The sample targeted urban precincts through geographic micro-targeting, focusing on areas where the party had either historically performed well or had recently lost significant ground to the green-left Mozemo party. The Social Democrats agreed to the experimental design due to practical constraints—they lacked the capacity to campaign across all precincts and needed to prioritize where to intervene. To guide this prioritization, k-means clustering was applied to past election results to identify clusters of strategic interest—areas where the party was currently performing well or had performed well in the past but had experienced recent decline.

Although both clustering and randomization were conducted at the district level, clustering served only as a targeting tool and played no role in the experimental assignment. Once relevant areas were identified, precincts within them were randomly assigned to treatment or control groups.

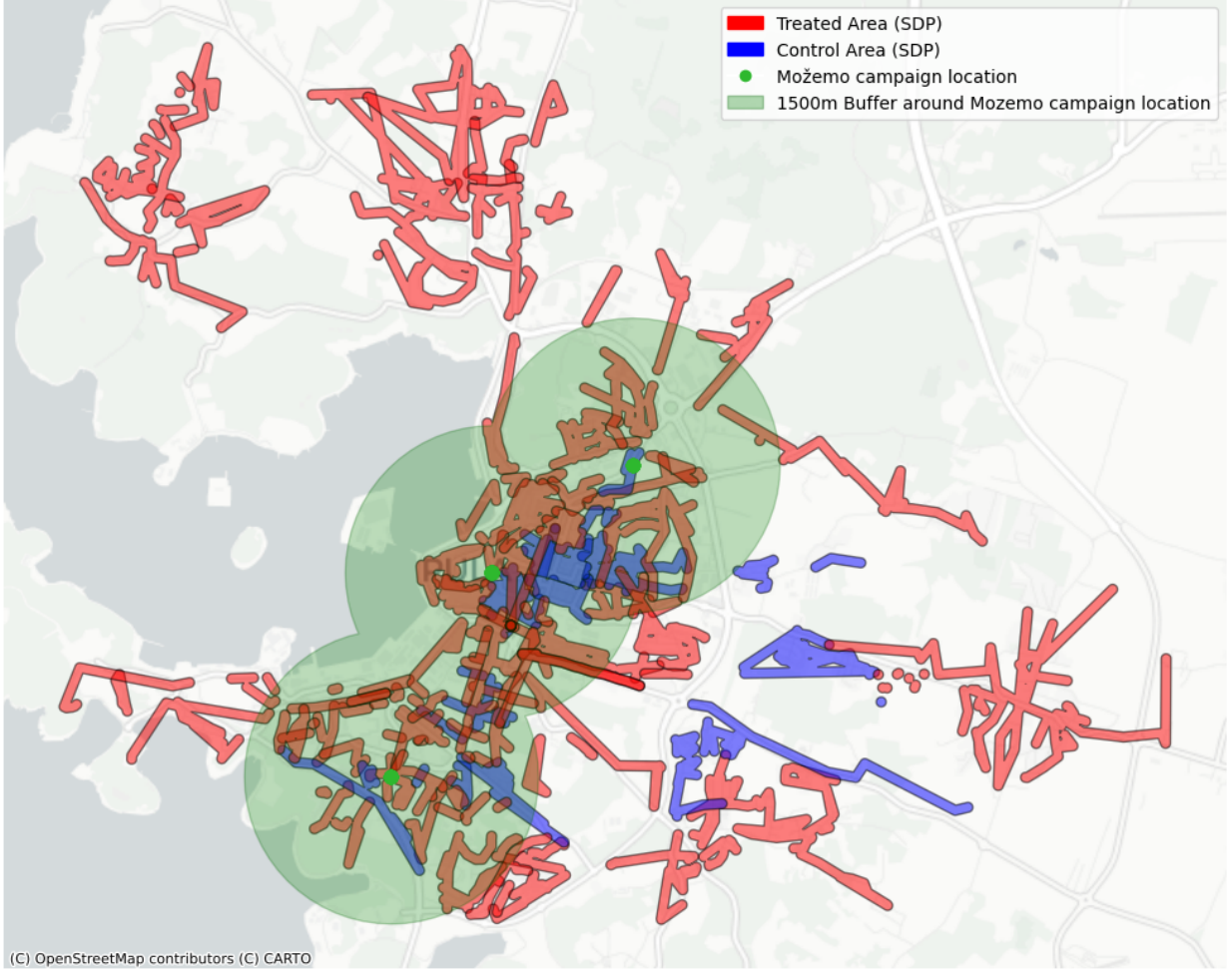


Figure 6: Precinct multipolygons and Možemo campaign locations included in the experiment within the administrative boundaries of the city of Pula.

9 Empirical Specifications

We aim to estimate the effect of SDP campaigning, conditional on Možemo’s activity, which was not randomized. One approach would be to include the full sample and estimate the interaction between SDP’s randomized treatment (“Treatment” in the equation below) and a binary indicator for whether Možemo campaigned in a given location—effectively a heterogeneity analysis.

$$Y_{ij} = \beta_0 + \beta_1 \text{Treatment}_i + \beta_2 \text{Možemo}_{ij} + \beta_3 (\text{Treatment}_{ij} \times \text{Možemo}_{ij}) + \gamma^\top \mathbf{X}_{2020,ij} + \delta_r + \epsilon_i \quad (1)$$

However, in this specification, only the interaction term can be interpreted causally, since SDP campaigning on its own is endogenous: it depends on Možemo’s prior decision to campaign in a location. On the other hand, Možemo campaigning alone versus competing against SDP

for the same voters is decided by the randomization in our experiment.

Furthermore, Gelman et al. (2020)¹⁶ shows that estimating interaction effects can require much larger sample sizes—up to 16 times larger in some cases—for the study to remain adequately powered. Given that expected effects are small (Coppock, Hill, and Vavreck 2020; Hewitt et al. 2024) and the aforementioned issues with power, this is likely too demanding for our experiment. However, we do report the results from this specification as an additional analysis. In our main specification, we restrict the analysis to a subsample of locations where (i) Mozemo campaigned and also (ii) which were also part of the randomization SDP engaged in. We then estimate a simple regression without an interaction term. This means we are estimating the effect of SDP campaigning in areas where Mozemo also campaigned. In this subsample, $\text{treatment} = 1$ indicates that both SDP and Mozemo campaigned in the location, and $\text{treatment} = 0$ indicates that only Mozemo did.

$$Y_{ij} = \alpha + \beta \cdot \text{Treatment}_{ij} + \gamma^\top \mathbf{X}_{2020,ij} + \delta_r + \varepsilon_{ij} \quad (2)$$

In this specification, Y_{ij} denotes one of four possible outcomes of interest in 2024 for unit i in town or city j : SDP vote share, Mozemo vote share, HDZ vote share, or turnout. Turnout is generally considered a key predictor of HDZ’s success: a truism in Croatian politics is that when overall turnout is low, the party still mobilizes its loyal base and the party performs relatively better¹⁷. The term Treatment_{ij} is an indicator for whether SDP campaigned in that unit (randomized). $\mathbf{X}_{2020,ij}$ represents a vector of covariates measured in 2020, including SDP vote share, HDZ vote share, turnout, other right-wing vote share, far-right vote share, and other left-wing vote share. δ_r denotes fixed effects for district r , and ε_{ij} is the error term, with standard errors clustered at the town/city level.

We define exposure based on proximity: our main specification uses a 1500-meter radius around Mozemo’s campaign locations. This is a realistic measure of campaign reach—roughly a 15–20 minute walk, which captures a local neighborhood. It is also the smallest radius that achieves covariate balance on pre-treatment vote shares and turnout. While the control and treatment units are balanced in terms of the pre-treatment covariates in the full sample, very small (nonrandom) subsets of the sample may not be. Small subsets around Mozemo’s location include the most central parts of a city, while increasing the radius encompasses more of the original full sample, leading to a balanced sample. The preferred specification

¹⁶see also the following blog post by Gelman for a discussion of this dynamic <https://statmodeling.stat.columbia.edu/2023/11/09/you-need-16-times-the-sample-size-to-estimate-an-interaction-than-to-estimate-a-main-effect-explained/>

¹⁷<https://www.jutarnji.hr/vijesti/hrvatska/otkad-je-hdz-prvi-put-izgubio-vlast-do-danas-vrijedi-isto-pravilo-i-opet-bi-moglo-presuditi-o-pobjedniku-izbora-15449602>

with a 1500-meter radius around Mozemo’s campaign locations reflects a tradeoff:

1. A larger radius ensures balance, as nearly all randomized neighborhoods are included
2. A very large radius (e.g., 5 km) is less realistically covered by Mozemo (it is not a priori sensible)
3. Voters at the outer edge of a large radius (like 5 km) are less likely to be exposed to Mozemo’s campaign which may lead to estimates that combine areas where effectively only SDP campaigned with those where both parties were truly active

We also report robustness checks using alternative radii of 1000m, 2000m, and 5000m. However, a 5000-meter radius in particular may be too broad, given the average size of Croatian cities, as it is roughly an hour’s walk. A very large radius such as 5000 m risks confounding the identifying variation by including areas far from the center of Mozemo’s activity. While some residents may use cars, many Croatian cities include pedestrianized areas.

10 Results

Table 1 presents balance on pre-treatment covariates across treatment and control groups within the 1500-meter radius subset. The means are similar across groups for all major 2020 vote share variables, including SDP, HDZ, Mozemo and other left-wing parties, far-right and other right-wing parties, as well as overall turnout.

	control	treatment	P_Value
SDP	0.333	0.327	0.502
HDZ	0.272	0.284	0.100
Mozemo and other left wing	0.155	0.149	0.382
DP (Far right)	0.076	0.081	0.120
MOST (Other right wing)	0.073	0.072	0.732
Other	0.069	0.066	0.277
Turnout	0.467	0.468	0.924

Table 1: Covariate Means by Treatment Status (1500m Radius Subset)

Table 2 shows post-treatment outcomes in 2024 by treatment status. Vote shares for SDP, Mozemo, and other major parties are comparable across groups, with a small but statistically significant increase in HDZ support in treated areas. Turnout is slightly lower in the treated group.

Table 3 presents results from the specification without covariate adjustment, using the full 1500m radius sample and robust standard errors clustered at the town level. The treatment

	control	treatment	P_Value
SDP	0.315	0.316	0.874
HDZ	0.249*	0.265*	0.026
Mozemo and other left wing	0.114	0.111	0.658
DP (Far Right)	0.070	0.073	0.311
MOST (Other right wing)	0.067	0.070	0.194
IDS	0.052	0.042	0.146
Fokus	0.023	0.019	0.064
Sjever	0.013	0.009	0.251
Turnout	0.640	0.633	0.195

Table 2: Covariate Means by Treatment Status (2024 Outcomes, 1500 m Radius Subset)

effect on HDZ vote share is positive and statistically significant, while the effects on SDP and Mozemo are near-zero and not statistically distinguishable from zero. Turnout effects are negative and consistently signed (across regressions), though significant at the 10% level in this specification.

Table 3: 1500 m radius subset, robust, clustered at town level, no controls)

<i>Dependent variable:</i>				
	SDP 2024	Mozemo 2024	HDZ (Center Right) 2024	Turnout 2024
	(1)	(2)	(3)	(4)
treatment	0.003 (0.005)	−0.001 (0.004)	0.010** (0.005)	−0.008* (0.005)
Constant	0.249*** (0.005)	0.207*** (0.007)	0.248*** (0.007)	0.699*** (0.005)
Observations	878	878	878	878
<i>Note:</i>			*p<0.1; **p<0.05; ***p<0.01	

Table 3 reports regression estimates using the full 1500m radius sample, now including covariates. The treatment effect on SDP vote share remains small and not statistically significant, suggesting the treatment did not help SDP gain more votes. The treatment increases HDZ vote share significantly ($p < 0.05$), while effects on Mozemo remains very close to zero and imprecise. While ensuring our sample is balanced is essential, we should also note that HDZ vote share is very strongly serially correlated (almost as much as turnout). This is likely due to the fact that voting for HDZ is habitual for those voters and they are often¹⁸ characterized as clientelistic (Kekez, 2018). Turnout is again negatively signed and, while

¹⁸see <https://www.politico.eu/article/croatias-ruling-conservatives-win-parliament-vote/>

not significant in this specific model, the magnitude is similar with some loss of precision, suggesting that reduced turnout may be the mechanism through which HDZ gains support. Also notable is that the estimated effect on HDZ vote share and on turnout are identical in absolute size – HDZ is estimated to gain 0.6 percentage points, while turnout declines by 0.6 percentage points. One interpretation is that left-wing voters, when reminded that SDP and Mozemo are competing separately may be more likely to abstain. This could be because the parties become less likely to win the election and win power so their vote is less likely to be pivotal, or because it indicates poor ability to reach deals that would allow them to govern, i.e. incompetence.

Table 4: 1500 m radius subset, robust and clustered at the town level, with controls

	<i>Dependent variable:</i>			
	SDP 2024 (1)	Mozemo 2024 (2)	HDZ (Center Right) 2024 (3)	Turnout 2024 (4)
treatment	0.005 (0.005)	0.001 (0.002)	0.006** (0.003)	−0.006 (0.004)
sdp_2020	0.103 (0.195)	0.014 (0.054)	−0.083 (0.077)	−0.204* (0.115)
hdz_2020	−0.126 (0.124)	0.026 (0.073)	0.635*** (0.083)	−0.168*** (0.062)
turnout_2020	0.109 (0.091)	0.035 (0.024)	0.019 (0.038)	0.762*** (0.056)
other_right_wing_2020	0.023 (0.168)	0.091 (0.088)	0.005 (0.111)	0.085 (0.111)
far_right_2020	−0.341** (0.140)	−0.084 (0.064)	0.124 (0.111)	−0.026 (0.084)
mozemo+_sdp_2020	0.077 (0.101)	0.605*** (0.052)	−0.155** (0.068)	0.008 (0.056)
Constant	0.212** (0.099)	0.020 (0.048)	0.110 (0.074)	0.395*** (0.067)
Observations	878	878	878	878

Note: *p<0.1; **p<0.05; ***p<0.01

Examining the basic specification for 2020 outcomes in table 5, as a placebo, yields

reassuring results. The 1500 m radius specification does *not* significantly predict any of the variables. While HDZ does have the same sign as in the main specification, it is not significant. Turnout is not significant and the coefficient is effectively 0.

Table 5: 1500 m radius subset, Placebo outcome

<i>Dependent variable:</i>				
	SDP 2020	Mozemo 2020	HDZ 2020	Turnout 2020
treatment	0.001 (0.005)	-0.004 (0.005)	0.005 (0.005)	-0.0003 (0.004)
Constant	0.226*** (0.005)	0.263*** (0.010)	0.277*** (0.007)	0.512*** (0.006)

Note:

*p<0.1; **p<0.05; ***p<0.01

We can then examine the other specifications – i.e. with varying radii around the Mozemo campaign locations. Tables 9-11 in the Appendix reveal that increasing the radius encompasses more of the units in the original randomization, leading to a more balanced sample with no significant differences. While the 1000 m specification has statistically significant imbalances, specifications with a larger radius rectify this issue.

Finally, we can examine the alternative specification discussed at the beginning of this section, i.e. corresponding to equation (1). Tables 7 and 8 examine the effects of SDP and Mozemo campaigning in the same area utilizing an interaction specification. Table 7, relying on a 1000 m radius around Mozemo locations (a priori perhaps preferable), indicates that the two parties competing for the same voters leads to a 1.1 percentage point decline in turnout, an effect size similar to the subset-based specifications. The effect on HDZ’s vote share is not significant in this specification, but it is directionally consistent with the other findings. Table 8 indicates similar findings with some loss of precision.

Table 6 presents a consistent pattern: there is no significant effect on either SDP or Mozemo vote share across specifications. In contrast, HDZ vote share indicates a persistently positive effect, with slightly smaller coefficients when controls are included, although the range of values remains stable across specifications. Turnout effects are generally negative but only statistically significant in the 1000m specification. It is likely that turnout is the channel through which HDZ gains a bigger vote share when Mozemo and SDP compete for the same voters.

This set of results suggests that parties that belong to the same ideological bloc competing

Table 6: Treatment Effects by Radius and Outcome

<i>Dependent variable:</i>				
	SDP 2020	Mozemo 2020	HDZ 2020	Turnout 2020
1000m radius				
treatment	0.003 (0.006)	−0.004 (0.004)	0.014** (0.005)	−0.011** (0.005)
Controls	No	No	No	No
Observations	747	747	747	747
treatment	0.006 (0.006)	0.0001 (0.003)	0.007** (0.003)	−0.009** (0.004)
Controls	Yes	Yes	Yes	Yes
Observations	747	747	747	747
2000m radius				
treatment	0.005 (0.004)	−0.0003 (0.003)	0.009** (0.004)	−0.002 (0.004)
Controls	No	No	No	No
Observations	1086	1086	1086	1086
treatment	0.006 (0.004)	0.0003 (0.002)	0.005* (0.002)	−0.003 (0.005)
Controls	Yes	Yes	Yes	Yes
Observations	1086	1086	1086	1086
5000m radius				
treatment	0.004 (0.004)	−0.001 (0.003)	0.010*** (0.004)	−0.002 (0.004)
Controls	No	No	No	No
Observations	1219	1219	1219	1219
treatment	0.005 (0.004)	−0.0004 (0.002)	0.005** (0.002)	−0.003 (0.004)
Controls	Yes	Yes	Yes	Yes
Observations	1219	1219	1219	1219

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 7: Regression Group (Full Sample, 1000 m radius around Mozemo, robust, no controls)

<i>Dependent variable:</i>				
	SDP 2024	Mozemo 2024	HDZ (Center Right) 2024	Turnout 2024
	(1)	(2)	(3)	(4)
treatment	−0.001 (0.004)	0.001 (0.002)	0.004 (0.003)	0.003 (0.003)
mozemo_1000	−0.011* (0.006)	0.008** (0.003)	−0.001 (0.004)	0.012*** (0.004)
intersection_1000	0.008 (0.007)	−0.001 (0.004)	0.004 (0.004)	−0.011** (0.005)
Controls	Yes	Yes	Yes	Yes
Observations	1666	1666	1666	1666

Note: *p<0.1; **p<0.05; ***p<0.01

Table 8: Regression Group (Full Sample, 1500 m radius around Mozemo, robust, no controls)

<i>Dependent variable:</i>				
	SDP 2024	Mozemo 2024	HDZ (Center Right) 2024	Turnout 2024
	(1)	(2)	(3)	(4)
treatment	−0.001 (0.005)	0.0002 (0.003)	0.004 (0.003)	0.002 (0.003)
mozemo_1500	−0.010* (0.006)	0.006* (0.003)	−0.0001 (0.004)	0.011*** (0.004)
intersection_1500	0.006 (0.007)	0.001 (0.004)	0.002 (0.004)	−0.008* (0.005)
Controls	Yes	Yes	Yes	Yes
Observations	1666	1666	1666	1666

Note: *p<0.1; **p<0.05; ***p<0.01

against each other may lead to cannibalization. As neither party gains in terms of vote share and turnout seems to drop among the bloc’s potential voters, parties outside the bloc mechanically benefit as their voters (unaffected by the intra-bloc competition) reliably turn out. Similar to the divided majority set-up in Cox (1997) and Bouton and Ogden (2018), a fragmented opposition that is unable to coordinate – i.e. form a pre-electoral coalition – may lead to the victory of the minority-preferred (but supported by partisan voters) candidate, that is, the Condorcet loser. While the left-wing parties in our setting are not necessarily the majority-preferred option, the strong opposition¹⁹ to the Christian Democrat HDZ across the political spectrum²⁰, including from far-right voters who perceived the ultimate coalition with HDZ as a betrayal²¹, suggests that HDZ may have in fact been a Condorcet loser in this election. Had the two left-wing parties formed a pre-electoral coalition, it would have been more likely for HDZ to lose power.

Our experimental setting can perhaps more directly speak to the effects of intra-bloc *competition* rather than the effects of coalitions, but we interpret the findings as an indicator of the costs of a lack of coordination, i.e. pre-electoral coalition formation. While we cannot ascertain whether the campaign effects are symmetric between coalition and non-coalition scenarios, our results suggest that, at a minimum, there are voters motivated to vote when the ideological bloc is likely to win, consistent with the basic intuitions of the pivotal voter model. In deciding whether to coordinate, parties may be particularly likely to resort to votes or polls among their members²². This prioritizes the preferences of perhaps their most partisan and ideologically driven voters, and sidelines the less attached voters. The more partisan members may be less likely to favor compromise through a coalition, while the party may not consider the marginal voters they could win through a coalition equally. Demonstrating the decline in turnout due to intra-bloc competition shows that, aside from a potential demotivating effect on expressive voters (i.e., partisans opposing a pre-electoral coalition) that parties may worry about in a coalition setting, there is also a loss of support in the *absence* of a coalition. If parties internalized this tradeoff, voters may find themselves better represented.

Furthermore, the canvassing literature generally indicates the practice produces small positive effects, the unusual result – i.e. falling turnout – suggests the context of the failed coalition is a significant one and that the success of the usual campaign persuasion tactics (which include canvassing but could perhaps also extend to the national-level media

¹⁹<https://www.vecernji.hr/vijesti/most-objavio-nositelje-lista-poslat-cemo-hdz-u-oporbu-1757010>

²⁰<https://www.index.hr/vijesti/clanak/glasaci-dpa-pisu-po-facebooku-stranke-samo-pregovaranje-s-hdzom-je-izdaja/2559238.aspx>

²¹<https://www.24sata.hr/news/nova-anketa-dvije-trecine-ljudi-smatra-kako-je-domovinski-pokret-prevario-svoje-birace-981933>*google_vignette*

²²see <https://www.nrc.nl/nieuws/2025/03/13/groenlinks-en-pvda-gaan-in-juni-al-stemmen-over-fusie-beslismoment-met-een-jaar-vervroegd-vanwege-rommelend-kabinet-a4886252>

appearances) depend on the strategic environment in the election.

11 Spillovers

We can also examine how key variables of interest vary in areas closer to Mozemo's exact campaign locations, as opposed to further away from them. Areas within, for example, a 2000m radius around Mozemo could have plausibly been part of Mozemo's selection process (i.e. a 2000 m radius might roughly correspond to a "neighborhood," which is the unit Mozemo articulated their strategy based on – and then targeted the busiest area within it). However, voters within a 1000m radius are more likely to be actually treated, while those between 1000 m and 2000 m may not have been exposed to Mozemo's campaign to quite the same extent. Table 9 shows this for a range of specifications – comparing areas up to 1000 m versus 2000 m away from Mozemo's campaign location, then 1500 m versus 2000 m and finally, 1500 m versus 2500 m. The results are fairly consistent across specifications. In areas outside of the narrow radius surrounding Mozemo's campaign location, SDP's campaign is associated with roughly a 1 p.p (corresponding to just the treatment indicator) increase in turnout, compared to areas outside of the narrow radius where SDP did not campaign. On the other hand, in the areas within the (more) narrow radius around Mozemo's campaign location, SDP's campaign is associated with almost a 2 p.p. reduction in turnout. This implies that moving from an area less exposed to Mozemo to an area more exposed to Mozemo's campaign leads the SDP campaign to produce a -0.6 or -0.9 p.p. decline (depending on the specification) in turnout. Furthermore, within areas closer to Mozemo's campaign location (within the more narrow radius), SDP's campaign is associated with almost a 1 p.p. increase in HDZ's vote share, while in areas further from Mozemo's exact campaign location, there is no such effect – the coefficient is negative, small and not significant.

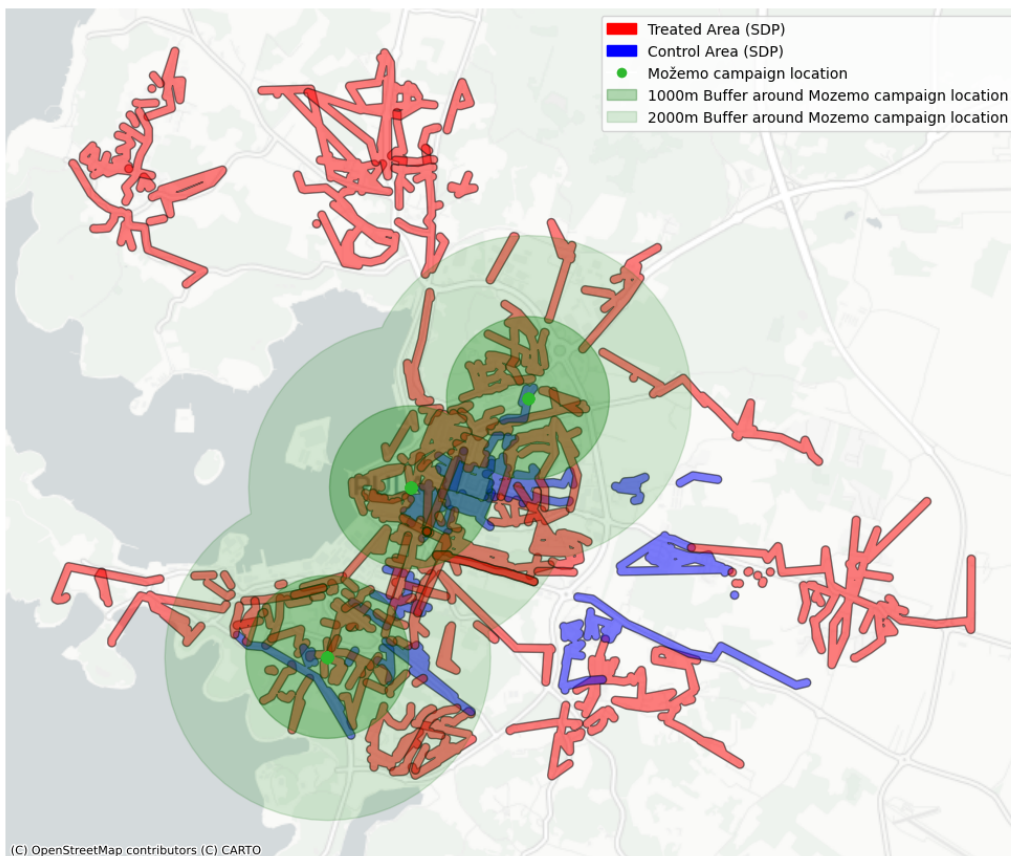


Figure 7: Visual illustration of the spillover specifications: 1000m and 2000m buffers around Mozemo campaign locations

Table 9: Spillovers at different radii

	<i>Dependent variable:</i>			
	SDP 2024	Mozemo 2024	HDZ (Center Right) 2024	Turnout 2024
1000m vs 2000m	(1)	(2)	(3)	(4)
treatment	0.004 (0.003)	0.0005 (0.003)	−0.002 (0.004)	0.010*** (0.004)
mozemo_1000	0.008 (0.005)	0.0002 (0.004)	−0.003 (0.004)	0.011** (0.005)
treatment:mozemo_1000	0.002 (0.005)	−0.0002 (0.004)	0.009* (0.005)	−0.019*** (0.005)
Observations	1086	1086	1086	1086
1500m vs 2000m	(1)	(2)	(3)	(4)
treatment	0.009** (0.004)	−0.004*** (0.001)	−0.003 (0.003)	0.012*** (0.003)
mozemo_1500	0.013 (0.009)	−0.009* (0.005)	0.001 (0.003)	0.008* (0.004)
treatment:mozemo_1500	−0.004 (0.006)	0.005* (0.003)	0.009** (0.004)	−0.018*** (0.004)
Observations	1086	1086	1086	1086
1500m vs 2500m	(1)	(2)	(3)	(4)
treatment	0.006* (0.003)	−0.005*** (0.002)	−0.003 (0.003)	0.011*** (0.004)
mozemo_1500	0.010 (0.007)	−0.007* (0.004)	−0.0003 (0.004)	0.006 (0.004)
treatment:mozemo_1500	−0.001 (0.005)	0.006** (0.003)	0.010** (0.004)	−0.017*** (0.005)
Observations	1115	1115	1115	1115
<i>Note:</i>				*p<0.1; **p<0.05; ***p<0.01

12 Conclusion

This study provides new experimental evidence on how voters respond when ideologically aligned parties fail to form a pre-electoral coalition. Using a large-scale randomized canvassing intervention during Croatia’s 2024 parliamentary elections, we show that when both the Social Democrats and the Greens campaign in the same area without having reached a coalition agreement, the main political beneficiary is the incumbent Christian Democrat party, likely due to a demobilization effect among left-leaning voters. These findings highlight a critical tension in fragmented multiparty systems: absent coordination, even programmatically similar parties may discourage their potential voters and unintentionally strengthen their ideological opponents. Our results underscore the strategic stakes of coalition-building in PR systems.

13 Bibliography

1. Abney, R., Adams, J., Clark, M., Easton, M., Ezrow, L., Kosmidis, S., & Neundorff, A. (2013). When does valence matter? Heightened valence effects for governing parties during election campaigns. *Party Politics*, 19(1), 61–82. <https://doi.org/10.1177/1354068811429623>
2. Abou-Chadi, T., & Kamphorst, J. (2023). The limits of issue entrepreneurship: How the German Greens failed to win in 2021. *Electoral Studies*, 85, 102532. <https://doi.org/10.1016/j.electstud.2023.102532>
3. Ansolabehere, S., & Iyengar, S. (1995). *Going negative: How political advertisements shrink and polarize the electorate*. New York: Free Press.
4. Arceneaux, K., & Nickerson, D. W. (2009). Who is mobilized to vote? A re-analysis of 11 field experiments. *American Journal of Political Science*, 53(1), 1–16. <https://doi.org/10.1111/j.1540-5907.2008.00354.x>
5. Bahnsen, O., Gschwend, T., & Stoetzer, L. F. (2020). How do coalition signals shape voting behavior? Revealing the mediating role of coalition expectations. *Electoral Studies*, 66, 102166. <https://doi.org/10.1016/j.electstud.2020.102166>
6. Barrie, C., et al. (2021). Tracking partisan sentiment using social media: Methodological challenges and new evidence. *PLOS ONE*, 16(12), e0259972. <https://doi.org/10.1371/journal.pone.0259972>
7. Barton, J., Castillo, M., & Petrie, R. (2014). What persuades voters? A field experiment on political campaigning. *The Economic Journal*, 124(574), F293–F326. <https://doi.org/10.1111/eoj.12093>
8. Bernhardt, D., Câmara, O., & Squintani, F. (2011). Competence and ideology. *The Review of Economic Studies*, 78(2), 487–522. <https://doi.org/10.1093/restud/rdq031>
9. Bergan, D. E., Gerber, A. S., Green, D. P., & Panagopoulos, C. (2005). Grassroots mobilization and voter turnout in 2004. *Public Opinion Quarterly*, 69(5), 760–777. <https://doi.org/10.1093/poq/nfi063>
10. Broockman, D. E., & Kalla, J. L. (2021). Which narrative strategies durably reduce prejudice? Evidence from field and survey experiments supporting the efficacy of perspective-getting. *American Journal of Political Science*, 66(1), 149–165. <https://doi.org/10.1111/ajps.12657>

11. Cantoni, E., & Pons, V. (2021). Do interactions with candidates increase voter support and participation? Experimental evidence from Italy. *Experimental Economics & Politics*, 33(2), 379–402. <https://doi.org/10.1111/ecpo.12167>
12. Coppock, A., Hill, S. J., & Vavreck, L. (2020). The small effects of political advertising are small regardless of context, message, sender, or receiver: Evidence from 59 real-time randomized experiments. *Science Advances*, 6(36), eabc4046. <https://doi.org/10.1126/sciadv.abc4046>
13. Cruz, C., Keefer, P., Labonne, J., & Trebbi, F. (2024). Making policies matter: Voter responses to campaign promises. *The Economic Journal*, 134(661), 1875–1913. <https://doi.org/10.1093/ej/uead042>
14. Dewan, T., Humphreys, M., & Rubenson, D. (2014). The elements of political persuasion: Content, charisma and cue. *The Economic Journal*, 124(574), F257–F292. <https://doi.org/10.1111/econj.12112>
15. Duch, R. M., May, J., & Armstrong, D. A. (2010). Coalition-directed voting in multiparty democracies. *American Political Science Review*, 104(4), 698–719. <https://doi.org/10.1017/S0003055410000420>
16. Enos, R. D., Fowler, A., & Vavreck, L. (2014). Increasing inequality: The effect of GOTV mobilization on the composition of the electorate. *The Journal of Politics*, 76(1), 273–288. <https://doi.org/10.1017/S0022381613001310>
17. Esteban Casanelles, T. (n.d.). Is an image worth a thousand votes? Evidence from a natural experiment in political advertising. Working paper.
18. Falcó-Gimeno, A., & Muñoz, J. (2017). Show me your friends: A survey experiment on the effect of coalition signals. *The Journal of Politics*, 79(4), 1454–1459. <https://doi.org/10.1086/693369>
19. Frey, A., López-Moctezuma, G., & Montero, S. (2021). Sleeping with the enemy: Effective representation under dynamic electoral competition. *American Journal of Political Science*, 66(3), 681–697. <https://doi.org/10.1111/ajps.12681>
20. Fujiwara, T., Müller, K., & Schwarz, C. (2021). The effect of social media on elections: Evidence from the United States. *NBER Working Paper No. 28849*. National Bureau of Economic Research. https://doi.org/10.3386/w28849/sol3/papers.cfm?abstract_id=3856816

21. Galasso, V., & Nannicini, T. (2011). Competing on good politicians. *American Political Science Review*, 105(1), 79–99. <https://doi.org/10.1017/S0003055410000523>
22. Galasso, V., Nannicini, T., & Nunnari, S. (2023). Positive spillovers from negative campaigning. *American Journal of Political Science*, 67(1), 5–21. <https://doi.org/10.1111/ajps.12610>
23. Gelman, A., Hill, J., & Vehtari, A. (2020). Regression and other stories. Cambridge University Press.
24. Gerber, A. S., & Green, D. P. (2000). The effects of canvassing, telephone calls, and direct mail on voter turnout: A field experiment. *American Political Science Review*, 94(3), 653–663. <https://doi.org/10.2307/2585837>
25. Gerber, A. S., & Green, D. P. (2001). Do phone calls increase voter turnout? An update. *Public Opinion Quarterly*, 65(1), 75–85. <https://doi.org/10.1086/320037>
26. Golder, S. N. (2006). Pre-electoral coalition formation in parliamentary democracies. *British Journal of Political Science*, 36(2), 193–212. <https://doi.org/10.1017/S0007123406000123>
27. Groseclose, T. (2001). A model of candidate location when one candidate has a valence advantage. *American Journal of Political Science*, 45(4), 862–886. <https://doi.org/10.2307/2669329>
28. Gschwend, T., & Hooghe, M. (2008). Should I stay or should I go? An experimental study on voter responses to pre-electoral coalitions. *European Journal of Political Research*, 47(5), 556–577. <https://doi.org/10.1111/j.1475-6765.2008.00775.x>
29. Gschwend, T., Meffert, M. F., & Stoetzer, L. (2017). Weighting parties and coalitions: How coalition signals influence voting behavior. *The Journal of Politics*, 79(2), 642–655. <https://doi.org/10.1086/688678>
30. Hewitt, A., et al. (2024). How experiments help campaigns persuade voters: Evidence from a large archive of campaigns’ own experiments. *American Political Science Review*, 118(1), 221–229. <https://doi.org/10.1017/S0003055423000879>
31. Hortalà-Vallve, R., Kanninen, V., & Rantapuska, E. (2023). Pre-electoral coalitions and voter behavior: Evidence from local elections. *Public Choice*, 197, 291–317. <https://doi.org/10.1007/s11127-023-01109-1>

32. Huber, G. A., & Arceneaux, K. (2007). Identifying the persuasive effects of presidential advertising. *American Journal of Political Science*, 51(4), 957–977. <https://doi.org/10.1111/j.1540-5907.2007.00291.x>
33. Ibenskas, R. (2015). Stability and survival of pre-electoral pacts: The role of incumbency status and party relations. *Party Politics*, 22(3), 352–362. <https://doi.org/10.1177/1354068814550437>
34. Kekez, A. (2018). Public service reforms and clientelism: Explaining variation of service delivery modes in Croatian social policy. *Policy and Society*, 37(3), 386–404. <https://doi.org/10.1080/14494035.2018.1436505>
35. Kendall, C., Nannicini, T., & Trebbi, F. (2015). How do voters respond to information? Evidence from a randomized campaign. *American Economic Review*, 105(1), 322–353. <https://doi.org/10.1257/aer.20131063>
36. Laver, M. (1998). Models of government formation. *Annual Review of Political Science*, 1, 1–25. <https://doi.org/10.1146/annurev.polisci.1.1.1>
37. Montero, J. (2024). Going it alone? A structural analysis of coalition formation in elections. *The Journal of Politics*, 87(2). <https://doi.org/10.1086/732989>
38. Nickerson, D. W., Friedrichs, R. D., & King, D. C. (2006). Partisan mobilization campaigns in the field: Results from a statewide turnout experiment in Michigan. *Political Research Quarterly*, 59(1), 85–97. <https://doi.org/10.1177/106591290605900107>
39. Pons, V. (2018). Will a five-minute discussion change your mind? A countrywide experiment on voter choice in France. *American Economic Review*, 108(6), 1322–1363. <https://doi.org/10.1257/aer.20160524>
40. Spenkuch, J. L., & Toniatti, G. (2018). Political advertising and election outcomes. *Quarterly Journal of Economics*, 133(4), 1981–2036. <https://doi.org/10.1093/qje/qjy019>
41. Zur, L. (2021). Stuck in the middle: Ideology, valence and the electoral failures of centrist parties. *British Journal of Political Science*, 51(2), 706–723. <https://doi.org/10.1017/S0007123419000378>

14 Appendix

Variable (2020)	control	treatment
SDP	0.347	0.332
HDZ	0.264*	0.285*
Mozemo and other left wing	0.153	0.143
DP (Far right)	0.073*	0.08*
MOST (Other right wing)	0.071	0.071
Other	0.071	0.067
Turnout_2020	0.467	0.467

Table 10: Covariate Means by Treatment Status (1000m Radius Subset)

Variable (2020)	Control	Treatment
SDP	0.316	0.310
HDZ	0.273	0.284
Mozemo and other left wing	0.165	0.165
MOST (Other right wing)	0.076	0.074
DP (Far-right)	0.080	0.083
Turnout	0.470	0.473

Table 11: Covariate Means by Treatment Status (2000m Radius Subset)

Variable (2020)	Control	Treatment
SDP	0.312	0.309
HDZ	0.276	0.286
Možemo and other left wing	0.167	0.164
DP (Far right)	0.081	0.083
MOST (Other right wing)	0.075	0.074
Populist	0.000	0.000
Turnout	0.469	0.471

Table 12: Covariate Means by Treatment Status (5000m Radius Subset)